ANY FAMILY OF LOCAL SECTIONS GENERATES A SMALLEST SUBSHEAF OF MODULES CONTAINING THEM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The subpresheaf of modules generated by a family of local sections, consisting of the finite sums of multiples of the sections over opens on which they are defined. Constructed inside the counted proof at source lines 626-637. Provides the presheaf whose sheafification will be the required subsheaf.
- The criterion that the sheafification of a subpresheaf of a sheaf of modules is again a subsheaf. Invoked inside the counted proof at source lines 638-640) as a cross-chapter pointer. Shows that the sheafification produced by the proof is a subsheaf of the given sheaf.

Source reference. Original reference sheaves-lemma-sheafify-epi-mono: sources/stacks-modules/pinned-original-source.tex, source lines 640–640. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-sheafify-epi-mono, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Let I be a set. Let s_i, $i \in I$ be a collection of local sections of $\mathcal{F}$, i.e., $s_i \in \mathcal{F}(U_i)$ for some opens $U_i \subset X$. There exists a unique smallest subsheaf of $\mathcal{O}_X$-modules $\mathcal{G}$ such that each s_i corresponds to a local section of $\mathcal{G}$.*

Proof. Consider the subpresheaf of $\mathcal{O}_X$-modules defined by the rule

$$U \longmapsto \{\text{sums } \sum\nolimits_{i \in J} f_i(s_i|_U) \text{ where } J \text{ is finite, } U \subset U_i \text{ for } i \in J, \text{ and } f_i \in \mathcal{O}_X(U)\}$$

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Let $\mathcal{G}$ be the sheafification of this subpresheaf. This is a subsheaf of $\mathcal{F}$ by Sheaves, Lemma sheaves-lemma-sheafify-epi-mono. Since all the finite sums clearly have to be in $\mathcal{G}$ this is the smallest subsheaf as desired. $\square$